Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.250 Hz*
Freq. upper bound:	*5.000 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*33*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*38*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *43.6%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1231*	*0.05369*
Peak wave period T_P (s):	*2*	*2*
Energy wave period $T_{-1,0}$ (s):	*1.781*	*1.896*
Mean wave period $T_{0,1}$ (s):	*1.603*	*1.572*
Mean wave period $T_{0,2}$ (s):	*1.49*	*1.337*
Spectral width (Broadness):	*0.7463*	*0.8706*
Spectral width (Narrowness):	*0.3977*	*0.6185*

Time Domain Analysis

Overall amplitude reflection coefficient: *43.8%*

	Incident	Reflected
Number of waves:	1008	994
Mean wave height H_m (m):	*0.07889*	*0.03554*
Mean wave period T_m (s):	*1.684*	*1.707*
Sig. wave height H_s (m):	*0.122*	*0.05541*
$T_{H_{1/3}}$ (s):	*1.897*	*1.991*
H_{max} (m):	*0.2042*	*0.1203*
T_{Hmax} (s):	*1.962*	*1.93*
$H_{1/10}$ (m):	*0.1543*	*0.06798*
$H_{1/50}$ (m):	*0.1846*	*0.08134*
$H_{1/100}$ (m)	*0.1931*	*0.08813*
$H_{1/250}$ (m):	*0.2*	*0.09885*
$H_{0.1\%}$ (m):	*0.2042*	*Not enough data*
$H_{1\%}$ (m):	*0.1859*	*0.07812*
$H_{2\%}$ (m):	*0.17*	*0.07276*
$H_{10\%}$ (m)	*0.1327*	*0.05848*
Groupiness factor GF:	*1.093*	*1.013*
Skewness b1:	*0.7105*	*0.1128*
Kurtosis b2:	*3.437*	*2.928*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *44.8%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1339*	*0.05999*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.822*	*1.966*
Mean wave period $T_{0,1}$ (s):	*1.608*	*1.569*
Mean wave period $T_{0,2}$ (s):	*1.474*	*1.297*
Spectral width (Broadness):	*0.7763*	*0.8823*
Spectral width (Narrowness):	*0.4345*	*0.6805*

Time Domain Analysis

Overall amplitude reflection coefficient: *44.7%*

	Incident	Reflected
Number of waves:	1023	1022
Mean wave height H_m (m):	*0.08767*	*0.04006*
Mean wave period T_m (s):	*1.749*	*1.75*
Sig. wave height H_s (m):	*0.1358*	*0.06165*
$T_{H_{1/3}}$ (s):	*1.963*	*2.079*
H_{max} (m):	*0.2226*	*0.1114*
T_{Hmax} (s):	*1.622*	*1.836*
$H_{1/10}$ (m):	*0.1672*	*0.07501*
$H_{1/50}$ (m):	*0.1928*	*0.08919*
$H_{1/100}$ (m)	*0.1983*	*0.09522*
$H_{1/250}$ (m):	*0.2044*	*0.1033*
$H_{0.1\%}$ (m):	*0.2221*	*0.1112*
$H_{1\%}$ (m):	*0.1931*	*0.08645*
$H_{2\%}$ (m):	*0.1822*	*0.08137*
$H_{10\%}$ (m)	*0.1479*	*0.06635*
Groupiness factor GF:	*1.098*	*0.9542*
Skewness b1:	*0.8143*	*0.08016*
Kurtosis b2:	*3.508*	*2.732*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.227 Hz*
Freq. upper bound:	*4.531 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*42*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *46.3%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1462*	*0.06763*
Peak wave period T_P (s):	*2.207*	*2.207*
Energy wave period $T_{-1,0}$ (s):	*1.893*	*2.07*
Mean wave period $T_{0,1}$ (s):	*1.638*	*1.578*
Mean wave period $T_{0,2}$ (s):	*1.483*	*1.268*
Spectral width (Broadness):	*0.8063*	*0.8895*
Spectral width (Narrowness):	*0.4693*	*0.7416*

Time Domain Analysis

Overall amplitude reflection coefficient: *46.3%*

	Incident	Reflected
Number of waves:	1028	1020
Mean wave height H_m (m):	*0.09633*	*0.04597*
Mean wave period T_m (s):	*1.797*	*1.811*
Sig. wave height H_s (m):	*0.1485*	*0.06874*
$T_{H_{1/3}}$ (s):	*2.066*	*2.128*
H_{max} (m):	*0.2458*	*0.1148*
T_{Hmax} (s):	*1.793*	*2.124*
$H_{1/10}$ (m):	*0.1807*	*0.08085*
$H_{1/50}$ (m):	*0.1991*	*0.09297*
$H_{1/100}$ (m)	*0.2062*	*0.09823*
$H_{1/250}$ (m):	*0.2188*	*0.1042*
$H_{0.1\%}$ (m):	*0.2449*	*0.1147*
$H_{1\%}$ (m):	*0.196*	*0.09197*
$H_{2\%}$ (m):	*0.1904*	*0.08543*
$H_{10\%}$ (m)	*0.1641*	*0.07315*
Groupiness factor GF:	*1.056*	*0.9071*
Skewness b1:	*0.8356*	*0.04089*
Kurtosis b2:	*3.397*	*2.593*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment_data.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *47%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1554*	*0.07302*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.925*	*2.128*
Mean wave period $T_{0,1}$ (s):	*1.637*	*1.57*
Mean wave period $T_{0,2}$ (s):	*1.468*	*1.238*
Spectral width (Broadness):	*0.8163*	*0.8916*
Spectral width (Narrowness):	*0.4938*	*0.7805*

Time Domain Analysis

Overall amplitude reflection coefficient: *47%*

	Incident	Reflected
Number of waves:	1028	1031
Mean wave height H_m (m):	*0.1049*	*0.05046*
Mean wave period T_m (s):	*1.851*	*1.845*
Sig. wave height H_s (m):	*0.1599*	*0.07369*
$T_{H_{1/3}}$ (s):	*2.118*	*2.183*
H_{max} (m):	*0.2425*	*0.1087*
T_{Hmax} (s):	*1.661*	*2.195*
$H_{1/10}$ (m):	*0.1902*	*0.08749*
$H_{1/50}$ (m):	*0.2137*	*0.09876*
$H_{1/100}$ (m)	*0.22*	*0.1026*
$H_{1/250}$ (m):	*0.2265*	*0.1058*
$H_{0.1\%}$ (m):	*0.242*	*0.1087*
$H_{1\%}$ (m):	*0.2134*	*0.09747*
$H_{2\%}$ (m):	*0.2023*	*0.09333*
$H_{10\%}$ (m)	*0.1724*	*0.07802*
Groupiness factor GF:	*1.073*	*0.8724*
Skewness b1:	*0.9266*	*0.01499*
Kurtosis b2:	*3.512*	*2.558*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment.dat*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *47.4%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1594*	*0.0755*
Peak wave period T_P (s):	*2.286*	*2.37*
Energy wave period $T_{-1,0}$ (s):	*1.96*	*2.169*
Mean wave period $T_{0,1}$ (s):	*1.645*	*1.549*
Mean wave period $T_{0,2}$ (s):	*1.46*	*1.199*
Spectral width (Broadness):	*0.8348*	*0.8947*
Spectral width (Narrowness):	*0.518*	*0.8172*

Time Domain Analysis

Overall amplitude reflection coefficient: *47.5%*

	Incident	Reflected
Number of waves:	1027	1055
Mean wave height H_m (m):	*0.1101*	*0.05274*
Mean wave period T_m (s):	*1.914*	*1.864*
Sig. wave height H_s (m):	*0.1651*	*0.0766*
$T_{H_{1/3}}$ (s):	*2.175*	*2.268*
H_{max} (m):	*0.2494*	*0.1169*
T_{Hmax} (s):	*1.844*	*2.467*
$H_{1/10}$ (m):	*0.1957*	*0.08992*
$H_{1/50}$ (m):	*0.2221*	*0.102*
$H_{1/100}$ (m)	*0.2323*	*0.1063*
$H_{1/250}$ (m):	*0.2427*	*0.1115*
$H_{0.1\%}$ (m):	*0.2493*	*0.1167*
$H_{1\%}$ (m):	*0.2183*	*0.101*
$H_{2\%}$ (m):	*0.2078*	*0.09578*
$H_{10\%}$ (m)	*0.1742*	*0.08096*
Groupiness factor GF:	*1.058*	*0.8473*
Skewness b1:	*0.9521*	*-0.01066*
Kurtosis b2:	*3.486*	*2.502*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap (FFT) of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *36.8%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08351*	*0.03073*
Peak wave period T_P (s):	*1.524*	*1.524*
Energy wave period $T_{-1,0}$ (s):	*1.441*	*1.454*
Mean wave period $T_{0,1}$ (s):	*1.364*	*1.29*
Mean wave period $T_{0,2}$ (s):	*1.314*	*1.172*
Spectral width (Broadness):	*0.5987*	*0.7934*
Spectral width (Narrowness):	*0.2784*	*0.4608*

Time Domain Analysis

Overall amplitude reflection coefficient: *36.8%*

	Incident	**Reflected**
Number of waves:	971	1011
Mean wave height H_m (m):	*0.05247*	*0.01948*
Mean wave period T_m (s):	*1.351*	*1.297*
Sig. wave height H_s (m):	*0.08114*	*0.03076*
$T_{H_{1/3}}$ (s):	*1.492*	*1.539*
H_{max} (m):	*0.137*	*0.05191*
T_{Hmax} (s):	*1.465*	*1.487*
$H_{1/10}$ (m):	*0.09891*	*0.03828*
$H_{1/50}$ (m):	*0.1163*	*0.04533*
$H_{1/100}$ (m)	*0.1224*	*0.04754*
$H_{1/250}$ (m):	*0.1286*	*0.05017*
$H_{0.1\%}$ (m):	*Not enough data*	*0.05189*
$H_{1\%}$ (m):	*0.1132*	*0.04408*
$H_{2\%}$ (m):	*0.1076*	*0.04226*
$H_{10\%}$ (m)	*0.08544*	*0.03302*
Groupiness factor GF:	*0.9726*	*1.001*
Skewness b1:	*0.3335*	*0.2962*
Kurtosis b2:	*2.934*	*3.04*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-12 Emerged Berm 2% - Test 2.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.266 Hz*
Freq. upper bound:	*5.313 Hz*

Freq. Domain Analysis

Overlap (FFT):	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*30*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*36*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *41.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1087*	*0.04501*
Peak wave period T_P (s):	*1.73*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.622*	*1.695*
Mean wave period $T_{0,1}$ (s):	*1.499*	*1.445*
Mean wave period $T_{0,2}$ (s):	*1.418*	*1.267*
Spectral width (Broadness):	*0.6832*	*0.8366*
Spectral width (Narrowness):	*0.3412*	*0.5484*

Time Domain Analysis

Overall amplitude reflection coefficient: *41.2%*

	Incident	**Reflected**
Number of waves:	1027	1020
Mean wave height H_m (m):	*0.06841*	*0.02947*
Mean wave period T_m (s):	*1.526*	*1.536*
Sig. wave height H_s (m):	*0.1063*	*0.04699*
$T_{H_{1/3}}$ (s):	*1.686*	*1.769*
H_{max} (m):	*0.1958*	*0.07919*
T_{Hmax} (s):	*1.681*	*1.426*
$H_{1/10}$ (m):	*0.1317*	*0.0597*
$H_{1/50}$ (m):	*0.1555*	*0.07173*
$H_{1/100}$ (m)	*0.1646*	*0.0745*
$H_{1/250}$ (m):	*0.1743*	*0.07786*
$H_{0.1\%}$ (m):	*0.1952*	*0.07919*
$H_{1\%}$ (m):	*0.1554*	*0.07149*
$H_{2\%}$ (m):	*0.1397*	*0.06641*
$H_{10\%}$ (m)	*0.1128*	*0.05102*
Groupiness factor GF:	*1.051*	*1.09*
Skewness b1:	*0.5579*	*0.2032*
Kurtosis b2:	*3.197*	*3.319*

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